

MINGZHE LI

Financial Economics, Household Finance, Continuous Time
Shihezi University ◊ mingzheli@stu.shzu.edu.cn

WORKING PAPERS

On the Monotonicity and Concavity of Effective Risk Aversion (2026) *Available at SSRN 6911380*

Abstract: Apparent risk aversion has been extensively studied in the savings problem, but the corresponding portfolio choice object—effective risk aversion, which is the direct determinant of the risky portfolio share—has never been characterized for heterogeneous investors. For HARA utility functions with $k > 1$, apparent prudence is greater than k times apparent risk aversion if and only if the consumption function is concave. Under this condition, this paper establishes necessary and sufficient conditions for effective risk aversion to be increasing and concave in wealth. In the Merton model, where apparent prudence equals k times apparent risk aversion, these properties hold whenever income exceeds the subsistence consumption. The results resolve the tension: under a decreasing MPC, apparent risk aversion falls in wealth whereas effective risk aversion rises, the resolution lying in the distinction between the optimal level of investment in risky assets and the share.

WORK IN PROGRESS

Household Problem with Risk Heterogeneity: A Continuous-Time Framework (2026)

Abstract: The classical framework assumes that the evolution of household welfare is deterministic and does not respond to risk sources, thus failing to distinguish the impact of aggregate risk versus idiosyncratic risk on household financial decisions. This paper relaxes this assumption, allowing the cost functional to have a stochastic response to any risk source. Using the Malliavin derivative as a sensitivity measure, the generalized HJB equation is derived through the Itô-Wentzell-Kunita formula. Compared to the classical framework, the generalized HJB equation includes a new term characterizing the cross-effect between household endowment risk exposure and welfare sensitivity to risk sources. We identify when the classical framework holds: risk sources do not directly affect household endowments (stochastic discounting, preference shocks, bequest functions, etc.), or households have no risk exposure (the Huggett model in continuous time).

COMPUTATIONAL WORK

Julia Implementation of "Financial Frictions and the Wealth Distribution" (2024)

Abstract: Independent implementation and replication of the model in "Financial Frictions and the Wealth Distribution, Fernández-Villaverde et al. (2023)" using Julia.

Open-Source Contribution for Modernization of "Recursive Macroeconomic Theory" Codebase (2024) *Available at GitHub Repository*

Abstract: This repository contains Matlab solutions and updated functions from Ljungqvist & Sargent's textbook (2018, MIT) for compatibility with Matlab 2024+.

EDUCATION

Shihezi University

2024-Present

Master of Finance (M.F.)

COMPUTATIONAL METHODS

Julia: Neural Networks, Numerical Analysis & PDEs, Tensor Calculation, Finite Difference Method

Matlab: DSGE

LANGUAGES

Chinese, English

Last updated: June 10, 2026